

DOCUMENT 08

Sources, licensing and unresolved choices

Primary evidence ledger and the checks required before production bundling

1. Evidence and permission boundaries

Research was performed on 23 September 2026 using the primary sources listed below. App paths were inspected on public main; several upstream projects use main/master. An immutable app commit was not established. Re-pin all implementation references before modifying code or comparing releases.

The supplied images are user-provided conceptual references. They have not been treated as pixel-accurate labels, original model output or permission to publish a training dataset. No checkpoint was downloaded/executed for benchmarking and no repository changes were made.

License findings, not blanket legal clearance

Artifact	Verified statement / limitation	Required action
Current app / CTD	The project documents GPL-3.0 and CTD provenance. [S01]	Preserve applicable notices and source obligations; review modified distribution.
Ogkalu detector/segmenter	Model cards identify Apache-2.0. [S25, S26]	Verify exact checkpoint, runtime-code dependencies and packaging terms separately.
COO DBNet++	COO links a checkpoint and the DB codebase. A complete redistribution clearance for the exact artifact was not established here. [S22, S23]	Confirm weight license/provenance and permitted production use before bundling.
COO MTSv3	Repository points to CC BY-NC 4.0. [S32]	Do not assume an unrestricted commercial option.
COO ABCNetv2	Instructions distinguish academic use from commercial use and request contact for commercial use. [S33]	Obtain appropriate permission or exclude it from that deployment.
Hi-SAM	Repository license is Apache-2.0; model setup also needs SAM encoder weights. [S27, S29]	Check the complete checkpoint stack and data/weight terms, not only the repo license.
Manga109 / Manga109-s	Full dataset and commercial-use subset have different conditions, including redistribution restrictions. [S34]	Identify the actual source volumes/terms for any training or published artifacts.

Downloading a weight, converting it to ONNX, distributing it separately, or putting it behind an optional switch does not by itself settle its license. Review the exact planned distribution with appropriate licensing expertise where needed.

Artifact security proposal

Pin download URLs, file hashes, model version and expected tensor signatures. Evaluate unfamiliar Python checkpoints in an isolated environment. Do not package arbitrary executable checkpoint loaders into the desktop app. Track converter versions and the final exported-file hash.

2. Sources: application core

S01 Manga Cleaner: project overview and current model inventory

S02 Manga Cleaner: CTD adapter and output contracts

S03 Manga Cleaner: region geometry and grouping

S04 Manga Cleaner: balloon detection and adopted-text mask helpers

S05 Manga Cleaner: script and outside-bubble gate

S06 Manga Cleaner: mask fitting and separate ink mask

S07 Manga Cleaner: binary mask and dilation primitives

S08 Manga Cleaner: model applied mask, isolation and blending

S09 Manga Cleaner: LaMa engine

S10 Manga Cleaner: fill engine

S11 Manga Cleaner: compositor and outside-mask invariant

S12 Manga Cleaner: patches and provenance

S13 Manga Cleaner: project persistence

How these sources are used: to identify existing responsibilities and proposed change locations. Source comments can document intent or an earlier measurement without proving a behavior on the user's installed build. The report distinguishes those comments from measurements performed in this session.

3. Sources: interface, orchestration and upstream CTD

- S14 Manga Cleaner: API region and mask metadata
- S15 Manga Cleaner: run lifecycle and model preconditions
- S16 Manga Cleaner: rectangular region overlay and hit targets
- S17 Manga Cleaner: page layer stack
- S18 Manga Cleaner: toolbar and capability checks
- S19 Manga Cleaner: user-facing cleaning modes and defaults
- S20 CTD upstream: inference and optional undetected-mask refinement
- S21 CTD upstream: image-guided mask refinement
- S35 Manga Cleaner: growth, isolation and edit-margin constants

Implementation verification: trace all call sites in a pinned checkout, including denoise and manual operations. The listed paths are not a substitute for compiling, running regression tests or inspecting a real prepared mask.

Source provenance notes

The source ledger is supplied in machine-readable JSON in the package. Per-document source links point to the same primary references. No benchmark score in this package is represented as an experiment run against the user's app or images.

The PDF/Word copies preserve the user reference images only to explain the requested behavior. Do not publish those pages or add them to training material without the necessary rights.

4. Sources: detector and segmentation research

S22 COO: dataset, detection checkpoints and benchmark results

S23 DB/DBNet++: official implementation and installation

S24 DB: text-region polygon postprocessing

S25 ogkalu: text-and-bubble detector model card

S26 ogkalu: YOLOv8m comic text segmenter model card

S27 Hi-SAM: stroke segmentation, checkpoints and patch inference

S28 Hi-SAM paper: pixel, word, line and paragraph segmentation

S29 Hi-SAM: Apache-2.0 repository license

S30 PaddlePaddle: PP-OCrv6 medium ONNX detector

S31 OpenCV: dilation and structuring elements

S36 Hi-SAM: executable stroke segmentation demo

Interpretation rule: detection, stroke segmentation, word/paragraph region segmentation and inpainting are different tasks. A result or model name from one task does not establish accuracy on the others. Tests on scene text or Japanese manga do not establish Korean-color-webtoon performance.

5. Sources: permissions and final decision checkpoints

S32 COO MTSv3: installation and non-commercial license reference

S33 COO ABCNetv2: installation and commercial-use notice

S34 Manga109 / Manga109-s: official dataset terms

Choices still requiring product or engineering sign-off

O01-O04: mode names/default, compact-group behavior, resolved padding settings, and handling of outlines/shadows/glows. See the complete register in Document 01.

O05-O08: production language/style coverage, the exact SFX model, whether another segmenter is justified, and acceptable device costs. Decide from the benchmark, not model popularity.

O09-O10: explicit outside-bubble/resume policy and exact artifact permissions. Neither is implied by selecting a new mask shape.

Recommended next implementation baseline

Keep legacy unchanged. Add text_shape with refined CTD evidence and explicit source-pixel padding; render the backend's exact allowed support; enforce it at every engine/commit boundary; add an explicitly selected no-recognition all-text policy. Benchmark CTD + COO as the SFX extension and check Korean pages before promotion.

SAM-TS/Hi-SAM and the comic segmenter are mask-quality experiments. PP-OCRv6 is a reserve discovery experiment. Rich semantic classification, manga-ocr dependency, VLM review and large multi-model loading are not requirements for this feature.

What has not been decided or delivered

No final best-model claim, tuned threshold set, hardware budget, production license clearance, trained custom model, benchmark result on the sample, source-code patch, pull request, or app release is included. The delivered artifacts are research and implementation documents.

Definition of success: the new option finds more target lettering, produces an editable text-shaped footprint with a small explicit margin, cleans only within that footprint, and coexists with the original option.